

TenderHub Deployment Manual

Production Cloud Architecture, Deployment Blueprint & Operations Guide

OFFICIAL SPECIFICATION

APPLICATION ARCHITECTURE Monorepo: Backend API + Frontend SPA + Admin Panel	PRIMARY CLOUD PLATFORM Render / Railway / Cloudflare Pages
DATABASE & CACHE MongoDB Atlas (M0/M10) + Upstash Redis	STORAGE LAYER Dual Cloudflare R2 Buckets (Primary & Backup)

1. Production Topology & Service Boundaries

TenderHub is split into three decoupled components to ensure high availability, fast asset delivery, and operational security:

Service Name	Directory	Type	Production Host	Internal Port
tenderhub-api	backend/	Node.js / Express Web Service	Render / Railway / AWS EC2	Port 8000
tenderhub-web	frontend/	React 19 / Vite Static Client	Cloudflare Pages / Vercel / Render	Port 5173 / CDN
tenderhub-admin	admin-panel/	React 19 / Vite Admin Dashboard	Private Subdomain / Cloudflare Access	Port 5174 / CDN
tenderhub-worker	backend/src/workers/	BullMQ Asynchronous Background Worker	Background Worker / Daemon Process	Headless Playwright

2. Option A: 1-Click Deployment via Render Blueprint (render.yaml)

The repository root contains an automated deployment specification (`render.yaml`).

Deployment Steps:

- Push repository to GitHub: `git push origin main`
- Open [Render Dashboard](#) > Click **New +** > Select **Blueprint**.
- Connect repository: Render automatically detects `render.yaml` .
- Provide required secrets (`MONGO_URI` , `REDIS_URL` , `R2_*`).
- Click **Apply**. Render builds and launches both the Web Service and Static Site concurrently.

3. Terminal Commands for Generating Both PDF Manuals

Terminal Automation: You can regenerate either or both instruction manuals anytime directly from the terminal inside the `backend` directory.

1. Generate Disaster Recovery & Backup Manual (PDF): `npm run manual:disaster` # 2. Generate Production Deployment & Operations Manual (PDF): `npm run manual:deployment` # 3. Generate BOTH Official Manuals Simultaneously: `npm run manuals:all`

TenderHub Deployment Manual

Environment Variables, Security Hardening & Health Telemetry

CONFIGURATION & AUDIT

4. Production Environment Variables Reference

Variable Name	Service Scope	Production Value / Recommendation
PORT	Backend	8000 (or auto-assigned by cloud provider)
NODE_ENV	Backend	production
MONGO_URI	Backend	MongoDB Atlas Primary Connection String
SECONDARY_MONGO_URI	Backend	Automated Fallback Cluster (Instant Failover)
REDIS_URL	Backend / Worker	Upstash Serverless Redis or Redis Cloud URI
R2_ACCOUNT_ID	Backend	Primary Cloudflare Account ID (tenderhub)
BACKUP_R2_ACCOUNT_ID	Backend	Secondary Cloudflare Account ID (tenderhub-backup)
ADMIN_SECRET_KEY	Backend / Admin	32-character random string for Admin Panel auth
VITE_API_URL	Frontend / Admin	Public Backend HTTPS URL (e.g. https://api.tenderhub.in)

5. Post-Deployment Verification & Health Check Checklist

- Health Endpoint:** Test https://your-backend.com/health (returns 200 OK with uptime).
 - Database Connectivity:** Check logs for MongoDB connected successfully [PRIMARY DB] .
 - Redis & BullMQ Queues:** Check logs for Redis Connected Successfully .
 - Scheduler Confirmation:** Verify all 5 automated scraping slots (09:00, 10:00, 13:00, 15:00, 18:30) and 02:00 AM retention cron are active.
 - Verify Admin Panel Telemetry:** Log in to http://localhost:5174 or your admin subdomain, open Telemetry & Health and Backup & Recovery.

Engineering Support: Both manual PDFs (TenderHub_Disaster_Recovery_and_Backup_Manual.pdf and TenderHub_Production_Deployment_Manual.pdf) are archived in the docs/ folder of the repository.